Seat No./Enrollment No.: _____

GUJARAT TECHNOLOGICAL UNIVERSITY
DIPLOMA IN ENGINEERING – SEMESTER 4 – EXAMINATION – SUMMER 2026

Subject Code: DI04016021 **Date: 19/05/2026**

Subject Name: Cyber Security and Digital Forensics

Time: 10:30 AM TO 01:00 PM **Total Marks: 70**

Instructions :
1. **Attempt all questions.**
2. **Make Suitable assumptions wherever necessary**
3. **Figures to the right indicate full marks.**
4. **Use of simple calculators and non-programmable scientific calculators are permitted.**
5. **English version is authentic.**

Q.1 **પ્રશ્ન ૧**	**(a)**	What is the CIA Triad ?	**03**
	(અ)	CIA Triad એટલે શું ?	**૦૩**
	(b)	Explain Authentication, Authorization, and Accounting (AAA) ?	**04**
	(બ)	Authentication, Authorization, and Accounting સમજાવો.	**૦૪**
	(c)	Explain the Seven Layers of Cyber Security	**07**
	(ક)	Seven Layers of Cyber Security સમજાવો..	**૦૭**
		અથવા OR	
	(c)	What is Vulnerabilities, Threats, and Risks ?	**07**
	(ક)	Vulnerabilities, Threats, and Risks એટલે શું ?	**૦૭**
Q.2 **પ્રશ્ન ૨**	**(a)**	Explain Security Attacks/Threats.	**03**
	(અ)	Security Attacks/Threats સમજાવો..	**૦૩**
	(b)	Explain Viruses, Worms, Trojan Horses, Ransomware .	**04**
	(બ)	Viruses, Worms, Trojan Horses, Ransomware સમજાવો.	**૦૪**
	(c)	Explain types of Operational Technology attacks .	**07**
	(ક)	Operational Technology attacks ના પ્રકાર સમજાવો.	**૦૭**
		અથવા OR	
Q.2 **પ્રશ્ન ૨**	**(a)**	Explain Proxy Server & Need of Private Proxy ?	**03**
	(અ)	Proxy Server અને Need of Private Proxy સમજાવો.	**૦૩**
	(b)	Explain Types of IOT Attacks.	**04**
	(બ)	Types of IOT Attacks સમજાવો .	**૦૪**
	(c)	Explain Phishing , E-mail spoofing and Spyware	**07**
	(ક)	Phishing , Email spoofing and Spyware સમજાવો .	**૦૭**

Q.3 **પ્રશ્ન 3**	**(a)**	What is Cybercrime and Classification of Cyber Criminals ?	**03**
	(અ)	Cybercrime એટલે શું ? Classification of Cyber Criminals લખો .	**03**
	(b)	Write a short note on classification of cyber-crimes.	**04**
	(બ)	classification of cyber-crimes ની ટુકનોંધ લખો .	**૦૪**
	(c)	Explain Cyber fraud and Man in the middle attack.	**07**
	(ક)	Cyber fraud અને Man in the middle attack સમજાવો.	**૦૭**
		અથવા OR	
Q.3 **પ્રશ્ન 3**	**(a)**	Explain Email Bombing , Logic Bomb.	**03**
	(અ)	Email Bombing અને Logic Bomb સમજાવો .	**03**
	(b)	Explain Web Jacking , Data diddling.	**04**
	(બ)	Web Jacking અને Data diddling સમજાવો,	**૦૪**
	(c)	Explain Cyber terrorism , cyber spying , Social Engineering Attack and Online gambling.	**07**
	(ક)	terrorism , cyber spying , Social Engineering Attack અને Online gambling સમજાવો .	**૦૭**
Q.4 **પ્રશ્ન ૪**	**(a)**	Explain Credit Card Fraud and Software Piracy.	**03**
	(અ)	Credit Card Fraud અને Software Piracy સમજાવો .	**03**
	(b)	Explain Copyright infringement and Trademarks violations.	**04**
	(બ)	Copyright infringement અને Trademarks violations સમજાવો.	**૦૪**
	(c)	Explain Concept of Hacking and Types of Hackers.	**07**
	(ક)	Hacking ના કોનસેપ્ટ અને Hackers ના પ્રકાર જણાવો.	**૦૭**
		અથવા OR	
Q.4 **પ્રશ્ન ૪**	**(a)**	What is malware? Explain common types of malware.	**03**
	(અ)	malware એટલે શું ? કોમન ટાઈપ ના malware સમજાવો .	**03**
	(b)	Explain Warning Signs of an Infection of Malware and Explain Prevention & Removal Tools.	**04**
	(બ)	Warning Signs of an Infection of Malware અને Prevention & Removal Tools સમજાવો,	**૦૪**
	(c)	What are the common Cyberattacks?	**07**
	(ક)	common Cyber attacks કયા કયા છે તે જણાવો ?	**૦૭**
Q.5 **પ્રશ્ન ૫**	**(a)**	Explain Network Forensics , Database Forensics.	**03**
	(અ)	Network Forensics અને Database Forensics સમજાવો .	**03**
	(b)	Explain Software forensics , Email Forensics and Malware Forensics.	**04**

(બ) Software forensics ,Email Forensics અને Malware Forensics સમજાવો . 04

(c) How following Devices are useful for Examined in Digital Forensics Investigations ? 07
1)Computers and Laptops
2) Smart Devices
3) Device Memory
4) Network Devices and Servers
5)CCTV

(ક) નીચેના સાધનો કેવી રીતે Digital Forensics Investigationsમાં મદદરૂપ થાય છે ? 07

1)Computers and Laptops

2) Smart Devices

3) Device Memory

4) Network Devices and Servers

5)CCTV

અથવા
OR

Q.5 પ્રશ્ન ૫

(a) Write a short note on Digital Forensics. 03

(અ) Digital Forensics ઉપર ટુકનોંધ લખો . 03

(b) Write a short note on cyber crime laws. 04

(બ) cyber crime laws ઉપર ટુકનોંધ લખો . 04

(c) Explain Remote Administration Tool (RAT) and Protect System from RAT. 07

(ક) Remote Administration Tool (RAT) અને Protect System from RAT સમજાવો . 07

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